

temperature. The increase in carbon dioxide concentration is largely the result of burning fossil fuels in vehicles and power stations.

EFFECTS OF GLOBAL WARMING

There is consensus among climate scientists that global warming is anthropogenic (caused by human activity). International agreements such as the Kyoto Protocol represent efforts to curb greenhouse gas emissions. Rising temperatures will have drastic consequences—meltwater will raise the sea level and increase flooding, while extreme weather events are likely to increase.



THE PREDICTED RISE IN SEA LEVEL BY THE END OF THE CENTURY

EXTREME WEATHER

Global warming appears to be increasing the frequency and severity of hurricanes, as the warmer atmosphere injects more energy and moisture into the climate system.

EARTH'S ENERGY BUDGET

Earth receives 174 PW (petawatts), which means it receives energy at 174 trillion joules per second. About 30 percent reflects back into space. The rest warms the surface and atmosphere, which produce energy as infrared radiation. Overall, the system is balanced—but the balance shifts as the concentration of greenhouse gases increases.

